CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (EC)

May 2013

EC 708 Advanced Optical Communication

Date: 13.05.2013, Monday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Explain outside vapor phase oxidation technique for the fabrication of optical fiber. [07]

Q - 2 (a) Enlist the requirements for selecting optical fibers. Show how acceptance angle is related to the fiber numerical aperture and refractive indices for the fiber core and cladding. [05]

(b) With the help of schematic diagram explain any one structure of LED [07]

(c) Define external quantum efficiency [02]

OR

Q - 2 (a) Sketch and explain ray optics representation of skew rays traveling in step index optical fiber core and meridional ray optics representation of propagation mechanism in an ideal step index optical waveguide [07]

(b) Discuss modulation process in LASER diode. [04]

(c) What is population inversion? [03]

Q - 3 (a) A double hetrojunction InGaAsP light emitting diode emitting at a peak wavelength of 1550nm has radiative and nonradiative recombination times of 40 and 110 ns, respectively. The drive current is 45mA.Calculate bulk recombination lifetime and internal quantum efficiency [04]

(b) Explain the principle and construction of reach through APD (Avalanche photodiode) with a neat sketch. [05]

(c) Describe the operation and principle of p-i-n photodiode with help of necessary diagram. [05]

OR

Q - 3 (a) Compare Graded Index fiber and Step Index fiber. [04]

(b) Write a note on noise sources and signal to noise ratio of photo detector noise. [05]

(c) How avalanche multiplication noise works? Explain in detail [05]

SECTION – II

Q - 4 What is WDM? Explain key features and advantages of WDM. [07]

Q - 5 (a) Write a note on SONET [07]

(b) Draw block diagram of an optical link using a frequency modulated carrier and explain example of unguided optical communication system for terrestrial systems. [07]

OR

Q - 5 (a) Explain the detail about solition pulses and parameters [07]

(b) Describe Neodymium LASER sources with external modulator [07]

Q - 6 (a) Write short note on application of optical fiber system in public network. [07]

(b) How optical fiber used in military application? Explain in detail. [07]

OR

Q - 6(a) Write short note on Consumer and civil application of Optical fiber system [07]

(b) Describe atmospheric attenuation with respect to transmission parameters in unguided optical communication system. [07]

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